Jefferson
Philadelphia University + Thomas Jefferson University

Pathology of the Thyroid and Parathyroid

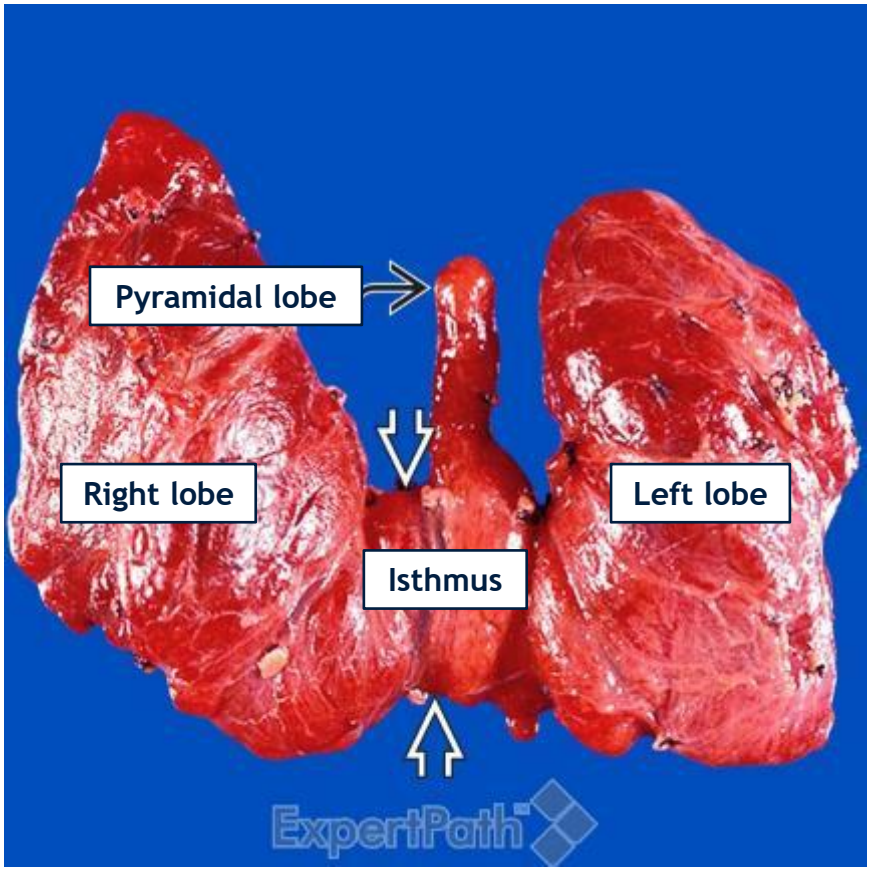
Stacey M. Gargano, MD (Stacey.Gargano@Jefferson.edu)
Associate Professor
Department of Pathology and Genomic Medicine

Learning Objectives

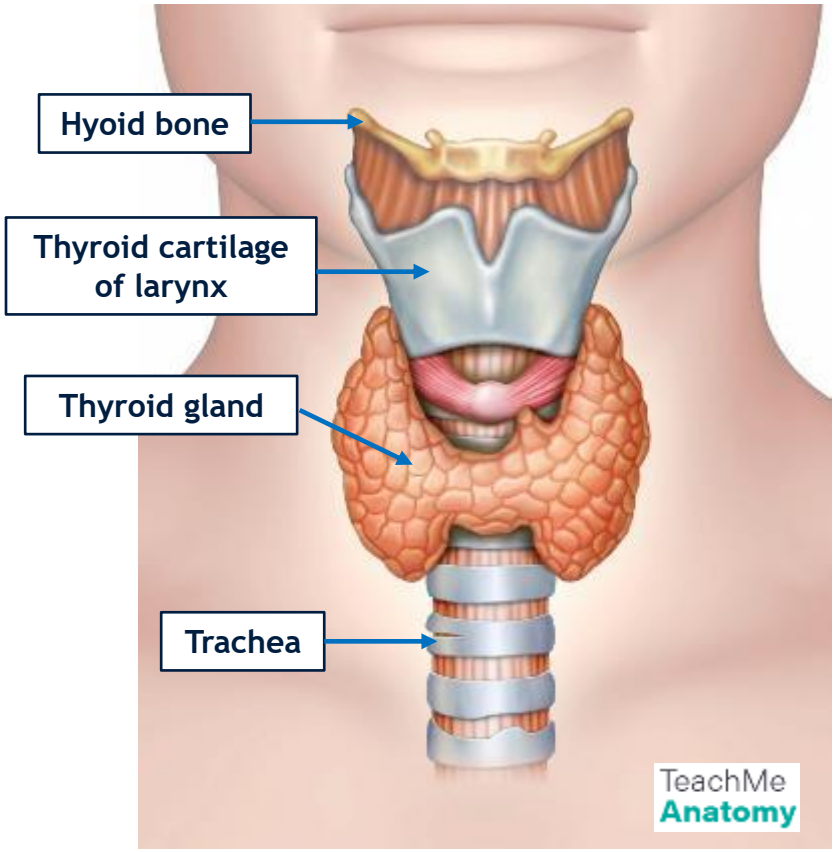
- Thyroid
  - Identify the main causes of hypo- and hyperthyroidism and their signs/symptoms
  - Describe and identify the histopathology of thyroid follicular nodular disease, Graves disease and Hashimoto thyroiditis
  - Explain the pathogenetic mechanisms underlying Graves disease and Hashimoto thyroiditis
  - Compare and contrast follicular adenoma, follicular carcinoma, papillary carcinoma, and medullary carcinoma
- Parathyroid
  - Distinguish between primary and secondary hyperparathyroidism based on clinical presentation, laboratory testing, and gross and microscopic pathological findings
  - Compare and contrast the histopathology of parathyroid hyperplasia and parathyroid adenomas

Normal Thyroid Anatomy and Function

Gross Anatomy

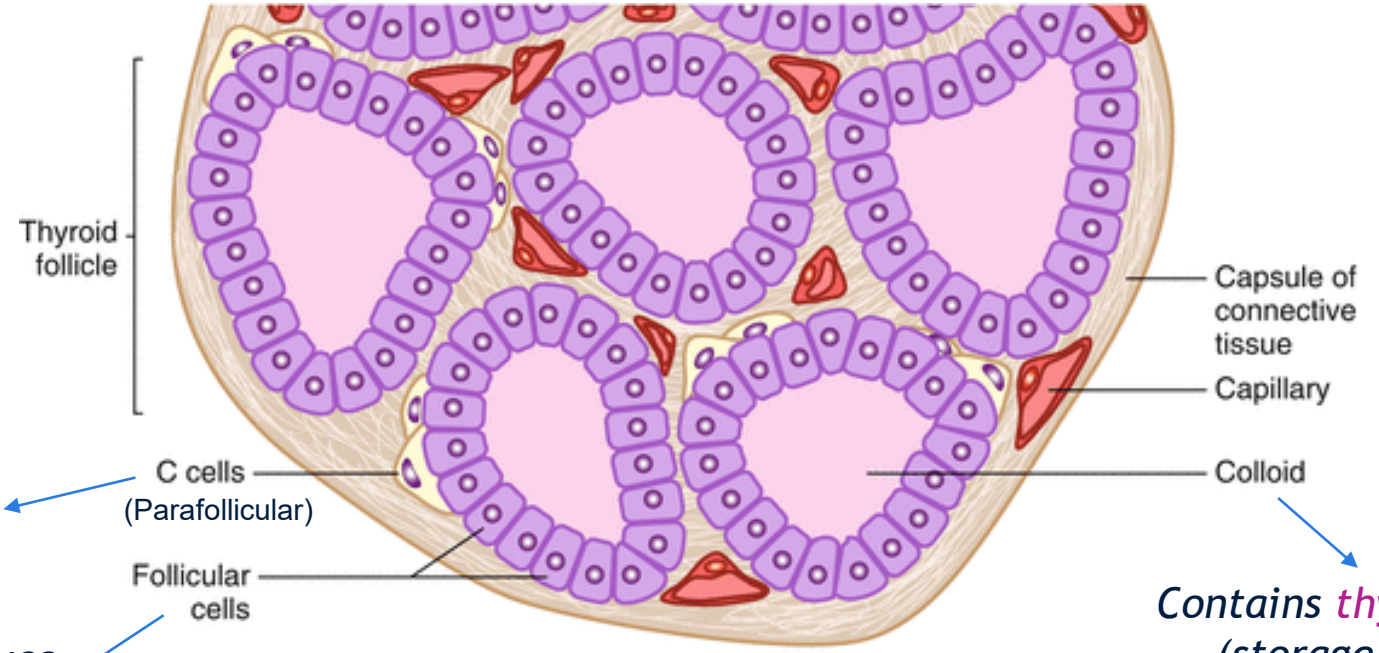


Anterior view



Microscopic Anatomy & Function

- **Follicle** = functional unit of the thyroid



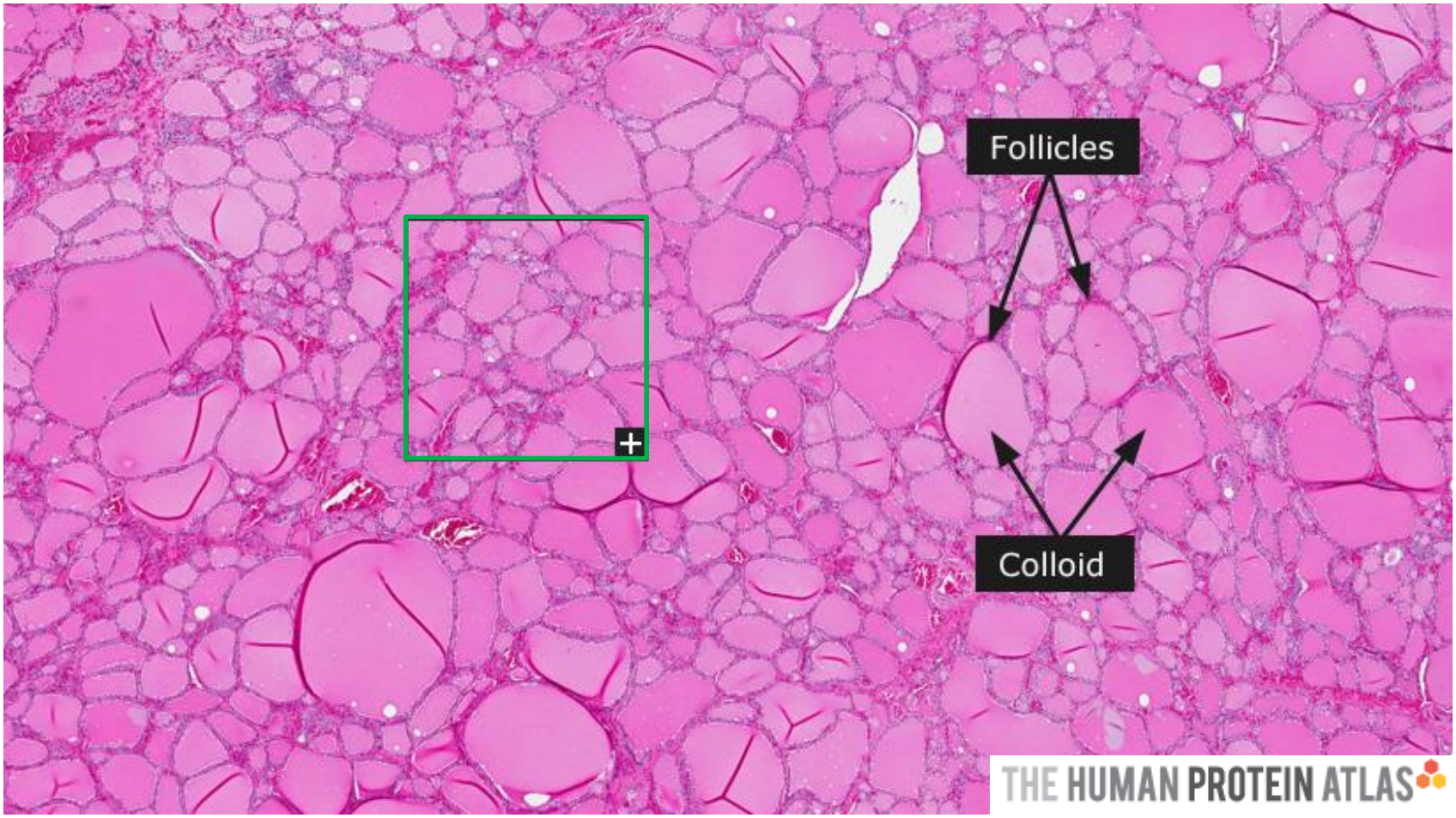
Produce calcitonin

Produce thyroid hormone

Contains thyroglobulin (storage form of thyroid hormone)

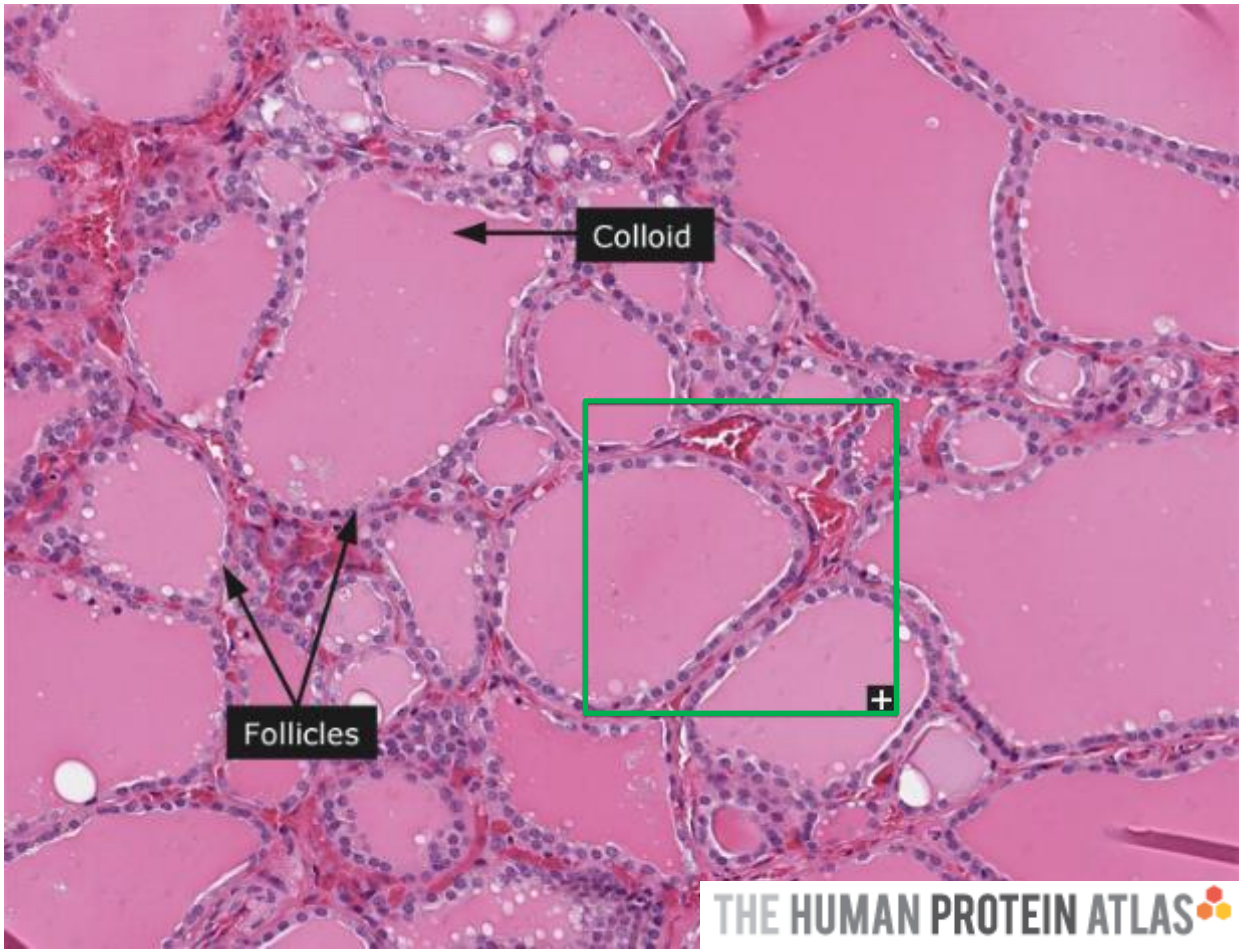
Rehfeld A., Nylander M., Karnov K. (2017) The Endocrine System. In: Compendium of Histology. Springer, Cham. https://doi.org/10.1007/978-3-319-41873-5_24

Histology (H&E)



Low power

Histology (H&E)



Intermediate power

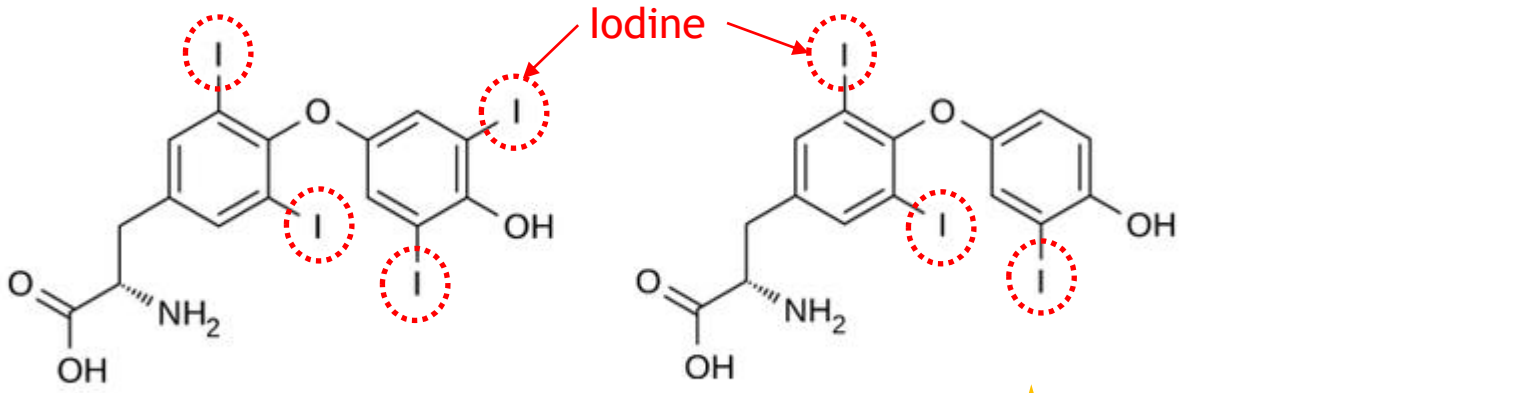
Histology (H&E)



High power

Thyroid Hormone

- Follicular cells secrete precursor glycoprotein called thyroglobulin (TG), which is stored in colloid
- In response to thyroid stimulating hormone (TSH) secreted from anterior pituitary, follicular cells convert TG into T3 and T4 *(catalyzed by thyroid peroxidase and requires iodine)* and release them into perifollicular capillaries

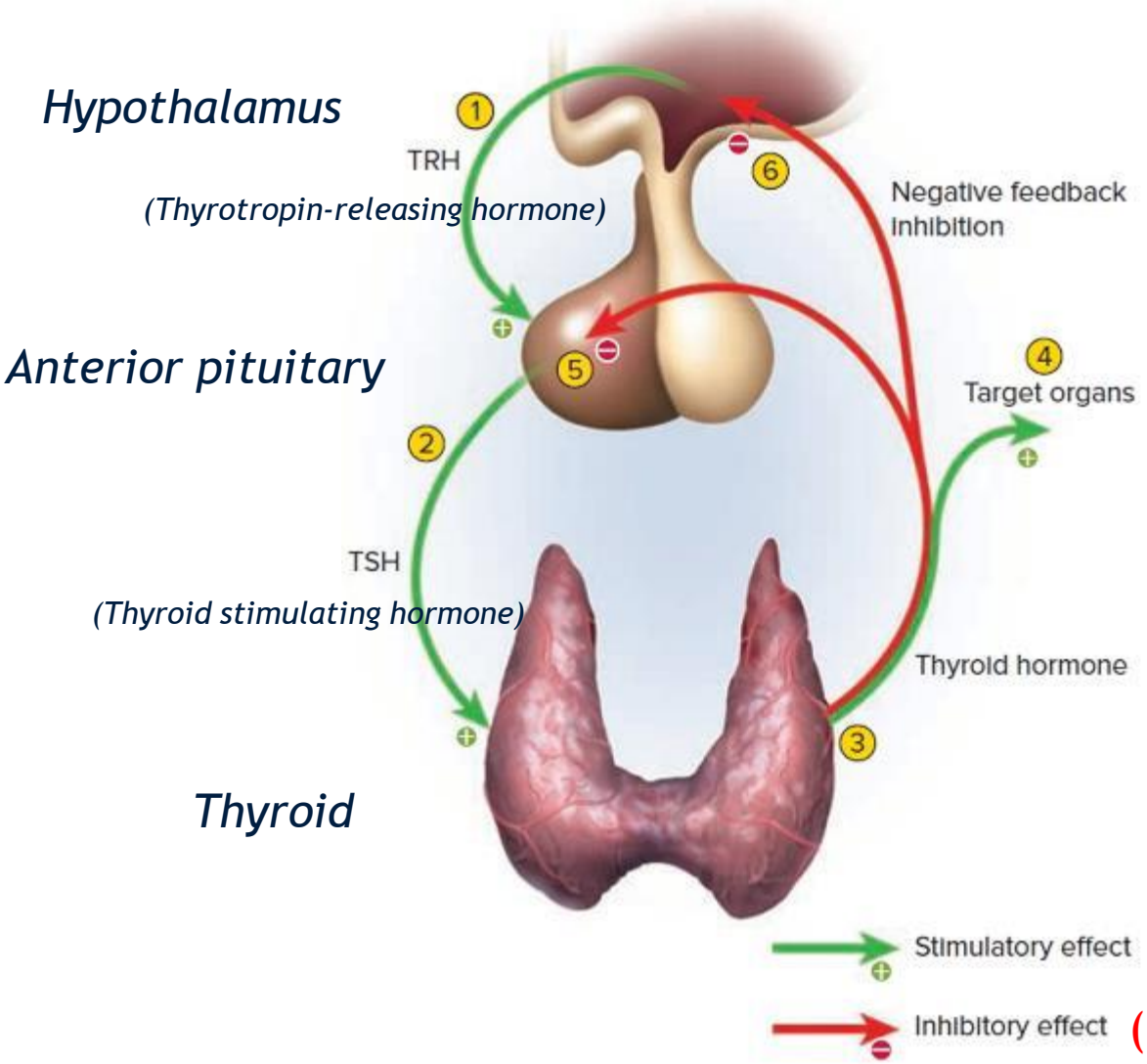


Thyroid Hormone Functions



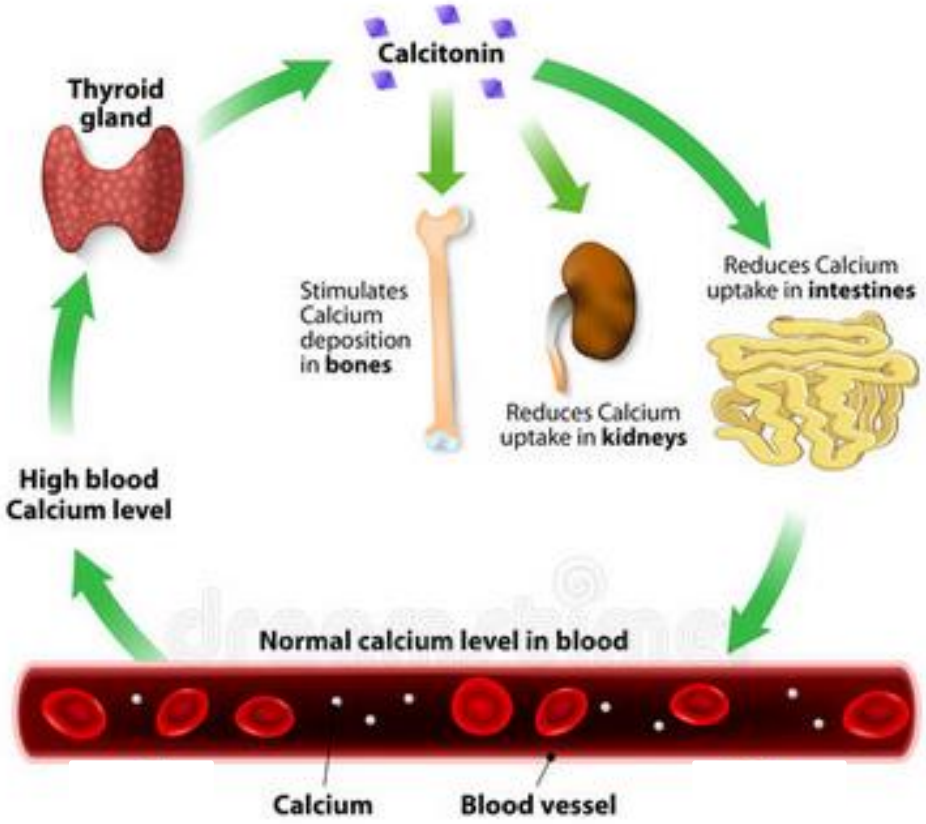
Metabolism regulation	Growth & development	Nervous system function
Digestive function	Temperature regulation	Heart rate & cardiovascular health
Muscle function	Reproductive health	Hair & nail health

Regulation of Thyroid Hormone Levels



(Feedback inhibition)

Calcitonin: Role in Calcium Homeostasis



Hyperthyroidism

Hyperthyroidism: Etiology

Primary

- Graves disease
- Thyroiditis
- Hyperfunctioning thyroid nodule(s)

Intrinsic abnormality of the thyroid

Secondary

- TSH-secreting pituitary adenoma

Overstimulation of thyroid by extrinsic source

Hyperthyroidism: Signs/Symptoms



Graves Disease

- #1 cause of hyperthyroidism
- Autoimmune disorder
  - Autoantibodies to TSH receptor lead to receptor *activation*
- Clinical triad:
  1. Thyrotoxicosis
  2. Exophthalmos
  3. Pretibial myxedema
- Labs:
  - ↑ serum T_3 and T_4, ↓ TSH

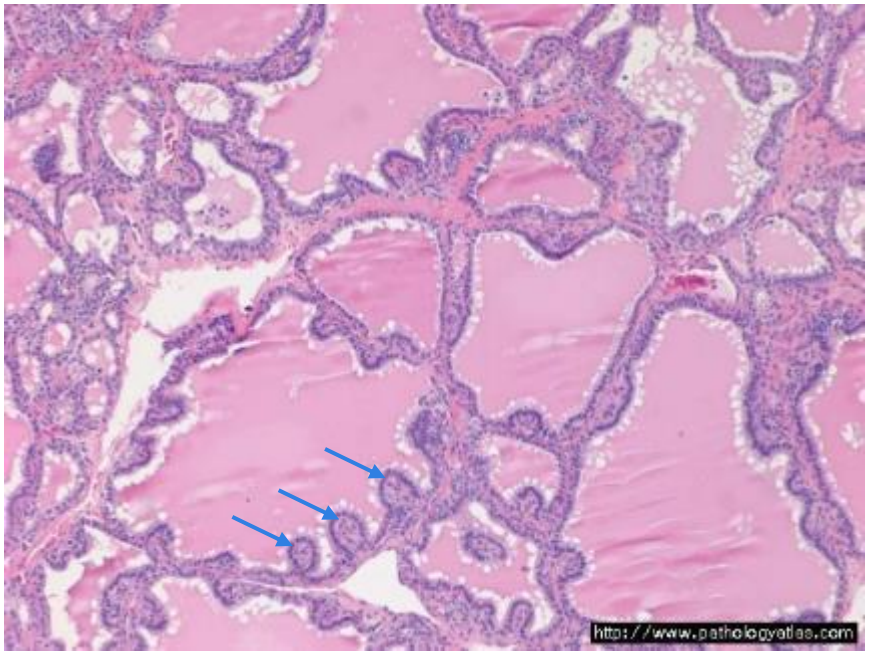




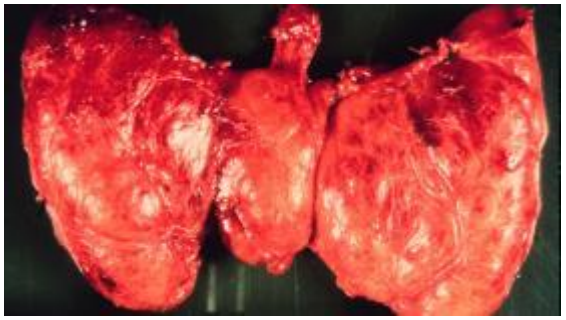
Fibroblasts in these locations express TSH receptor & secrete excess ECM proteins upon stimulation by autoAb's

Graves Disease: Pathology

= Diffuse hyperplasia



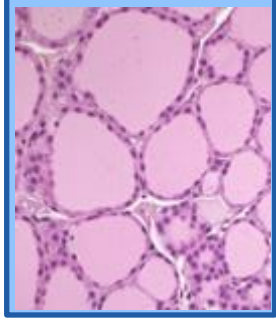
Variably sized follicles lined by hyperplastic follicular epithelium forming projections into lumen (arrows)

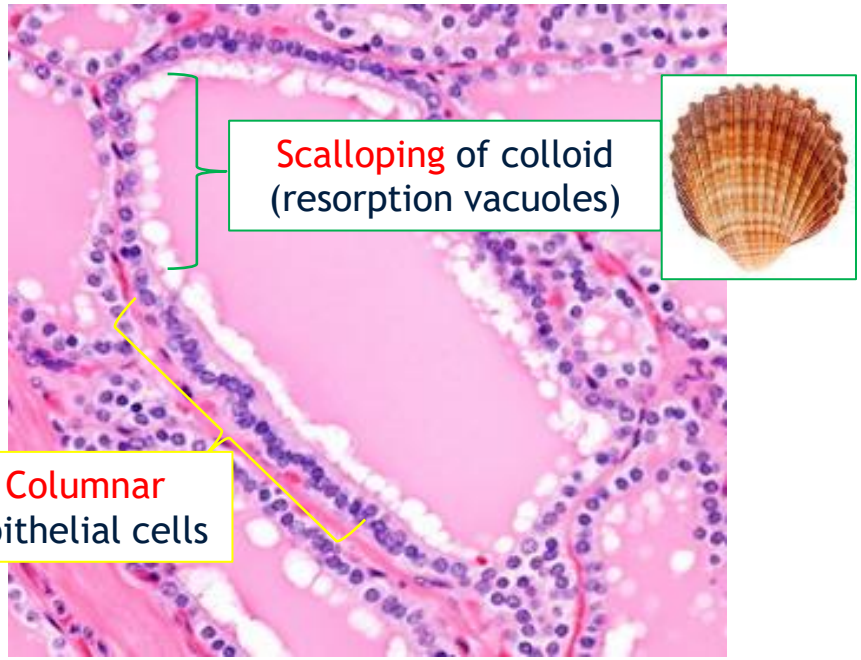


Diffuse symmetric enlargement of gland



Normal





Hypothyroidism

Hypothyroidism: Etiology

Primary

- Iodine deficiency
- Hashimoto thyroiditis

Defective synthesis of thyroid hormone

Secondary

- Damage to hypothalamus or pituitary by:
  - Surgery
  - Inflammation
  - Radiation therapy

Inadequate secretion of TSH

Hypothyroidism: Signs/Symptoms

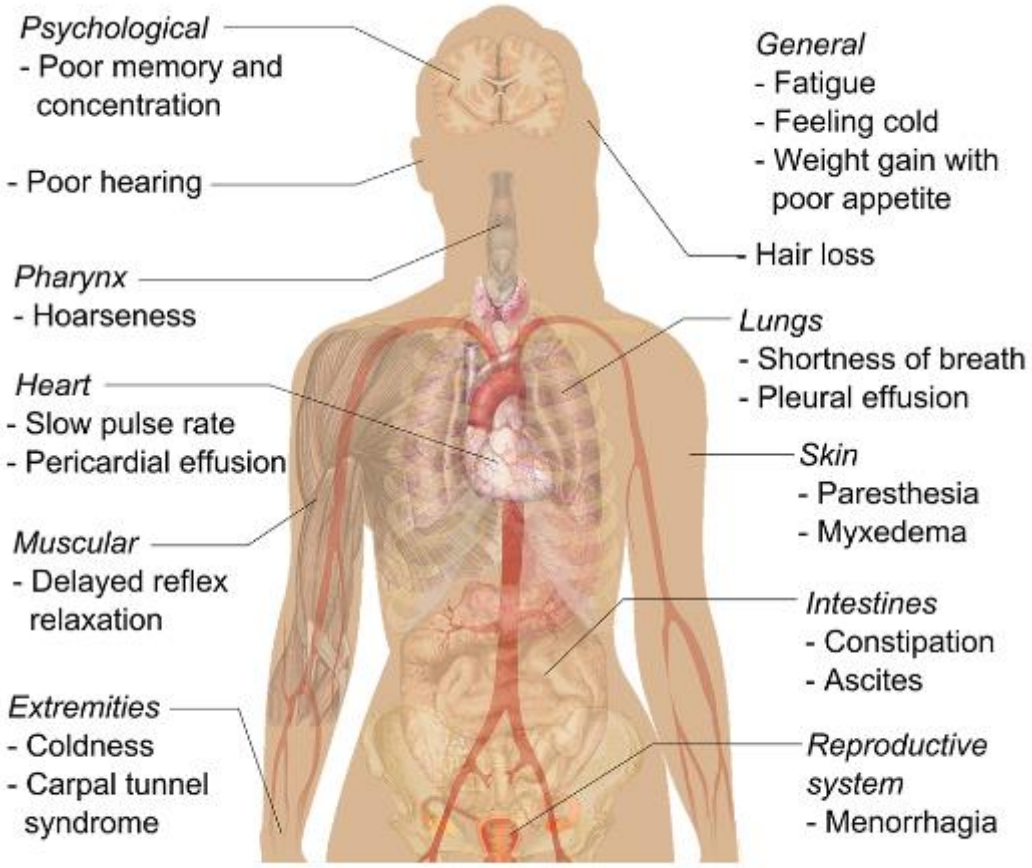


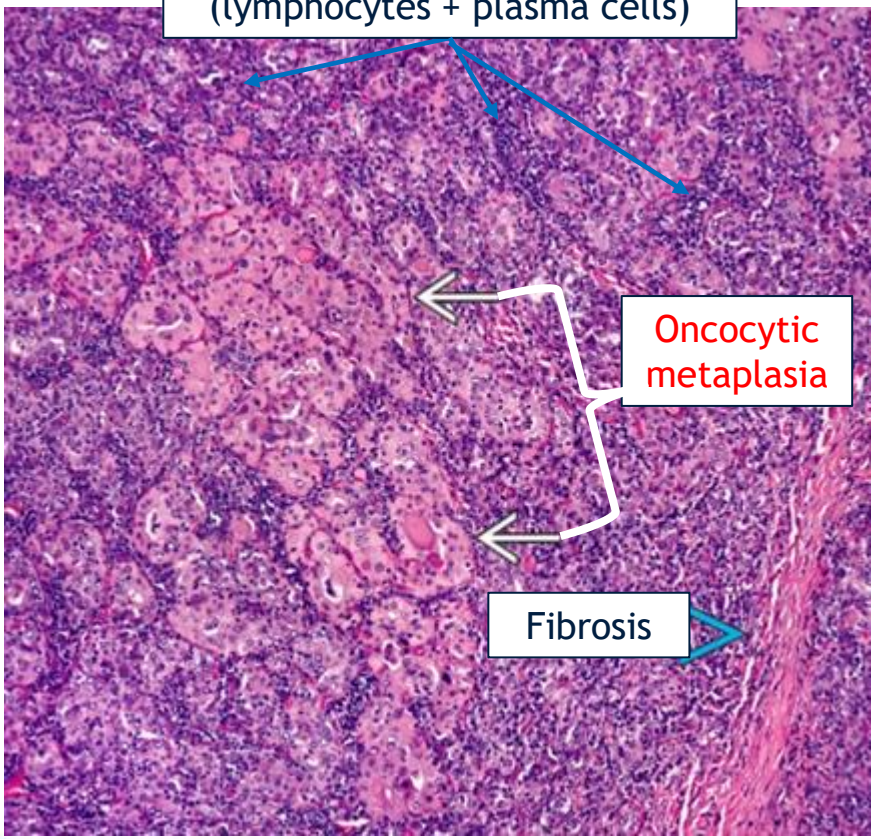
Image via: Wikipedia.org

Hashimoto Thyroiditis

- #1 cause of hypothyroidism *in regions where iodine levels are sufficient*
- Autoimmune disorder
  - Autoantibodies to thyroid peroxidase and thyroglobulin lead to *destruction* of thyroid parenchyma
- Labs:
  - $\downarrow T_4$, $\uparrow$ TSH
- Increased risk of papillary thyroid carcinoma and B-cell lymphoma of the thyroid

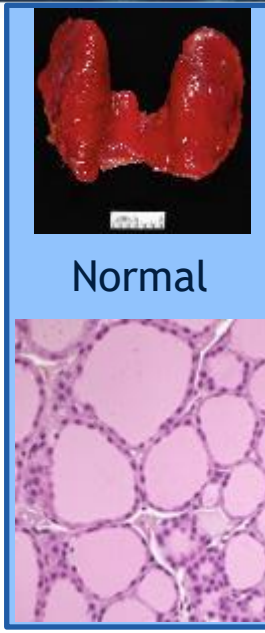
Hashimoto Thyroiditis: Pathology

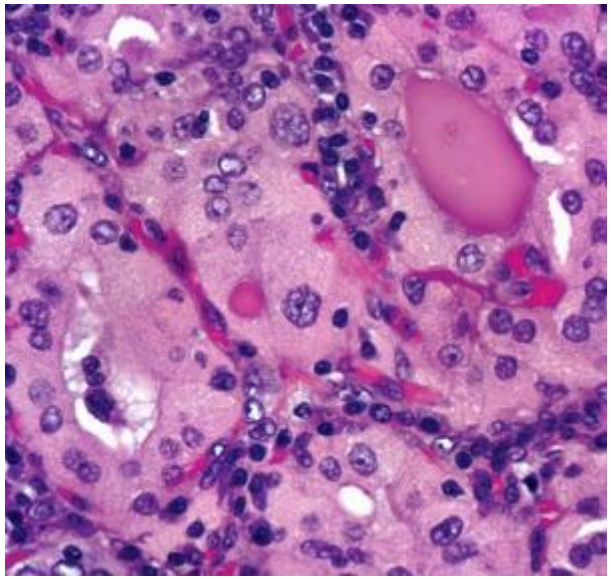
= Chronic lymphocytic thyroiditis





Enlarged gland with lobulated tan-white cut surface





Oncocytic cells (formerly Hürthle cells)

- Abundant granular, pink cytoplasm (due to ↑mitochondria)

Goiter

Goiter

- Enlargement of the thyroid
  - Diffuse or multinodular
  - Toxic or nontoxic

- Causes
  - Overstimulation of thyroid gland
    - Graves disease (→ diffuse toxic goiter)
  - Impaired synthesis of thyroid hormone
    - Hashimoto thyroiditis (→ diffuse or multinodular goiter)
    - Iodine deficiency
    - Drugs (e.g. lithium)
    - Goitrogens (e.g. turnips)
    - Hereditary factors

→ Multinodular goiter (MNG)



Patient with multinodular goiter

MNG: Pathology

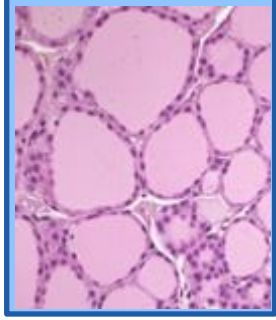
= Thyroid follicular nodular disease



Asymmetrically enlarged gland with nodularity

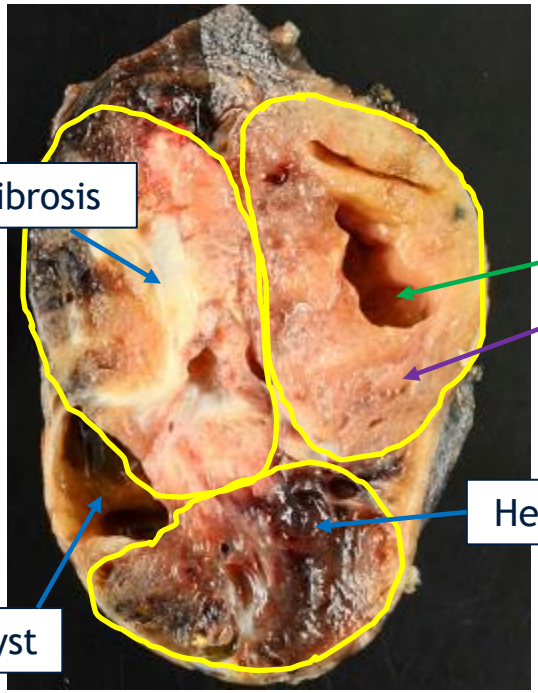


Normal





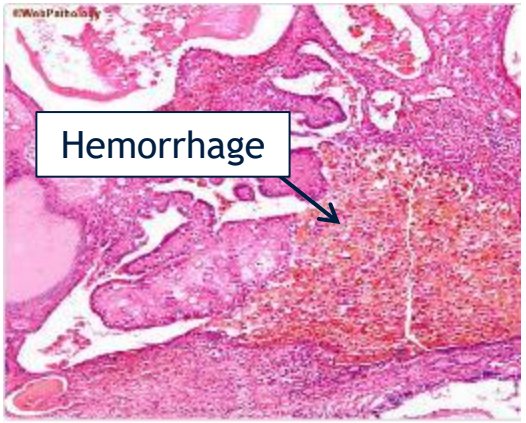
Multiple, variably sized nodules composed of follicles with varying degrees of cellularity and colloid



Cut surface with multiple nodules

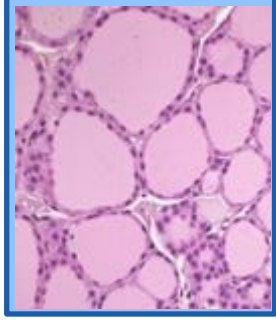
MNG: Pathology

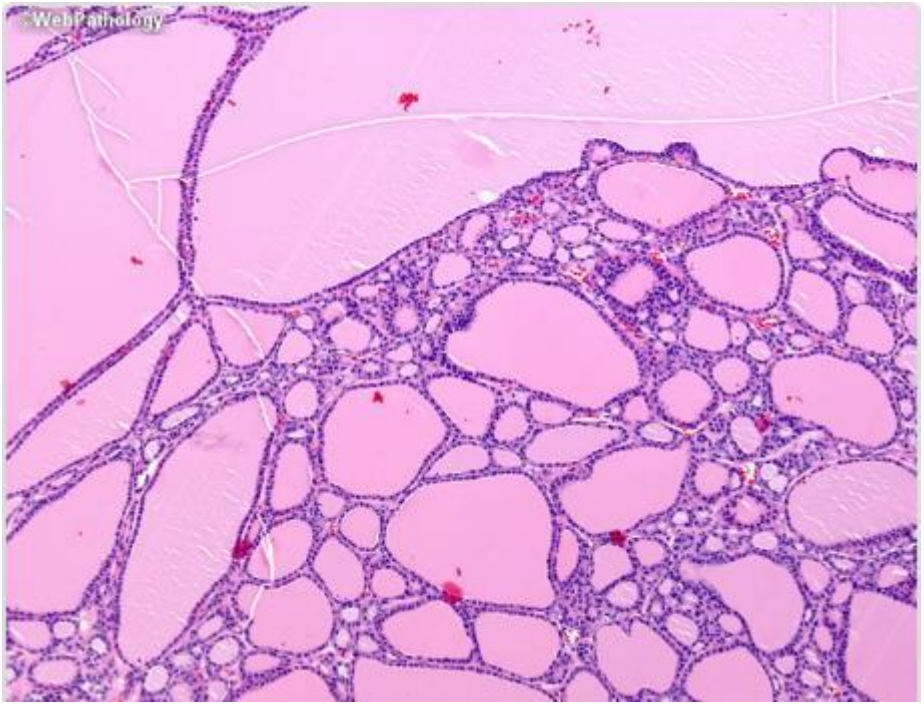
= Thyroid follicular nodular disease



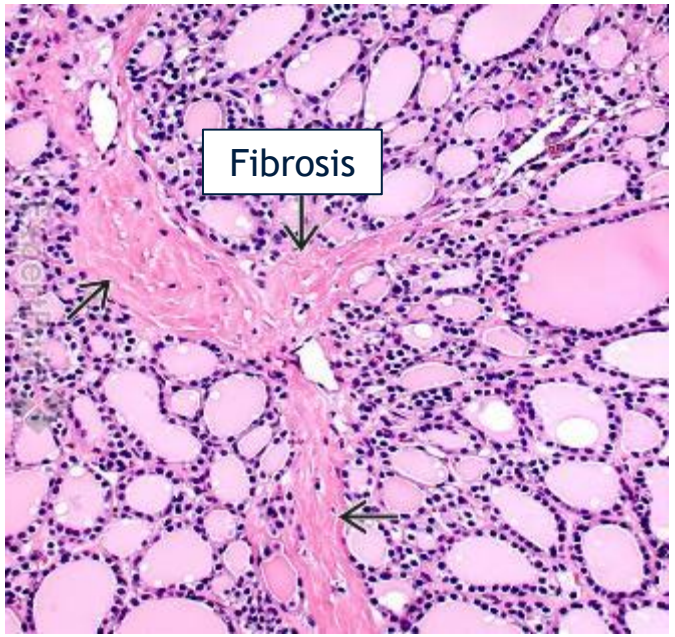


Normal





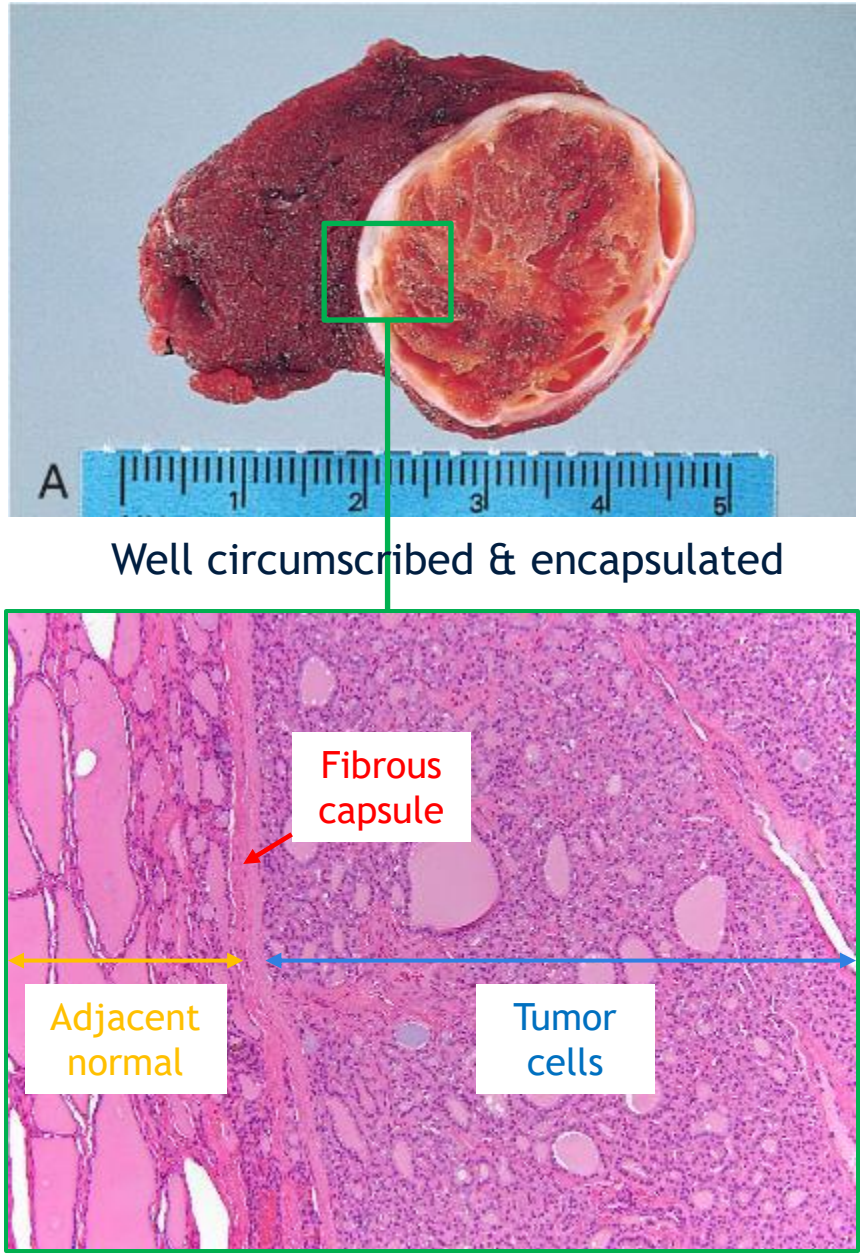
Mixture of small and large follicles



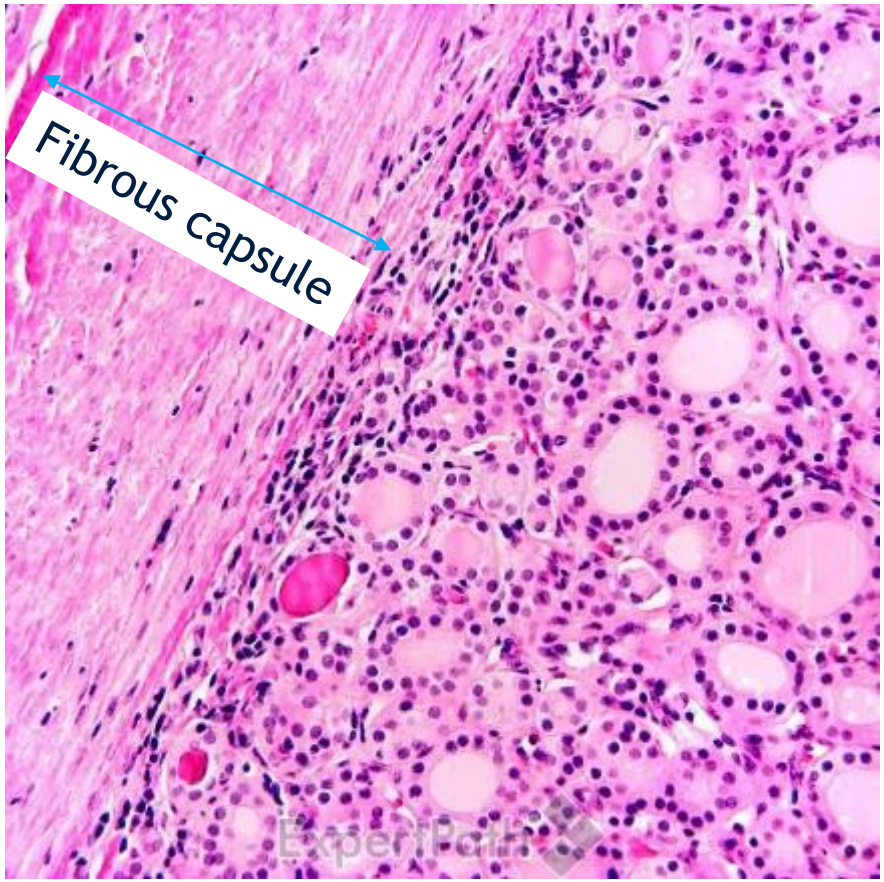
Thyroid Neoplasia

Follicular Adenoma

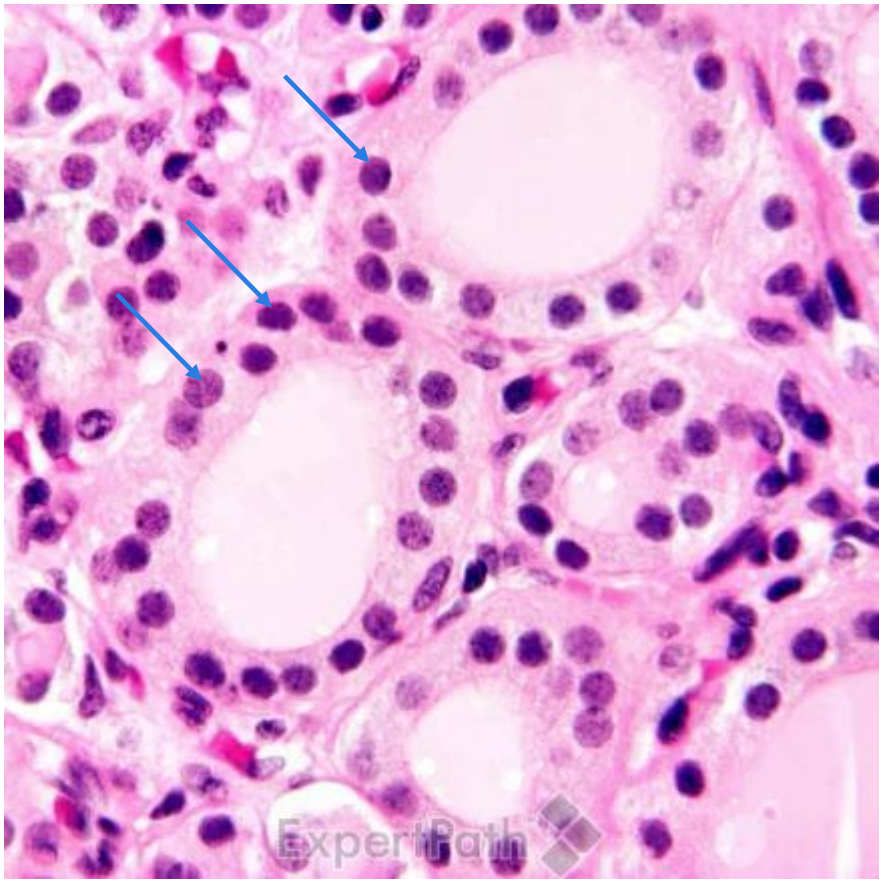
- #1 benign thyroid neoplasm
- Derived from follicular epithelium
- Majority are nonfunctional
  - Rarely produce thyroid hormones (*toxic adenomas*) → hyperthyroidism



Follicular Adenoma: Pathology

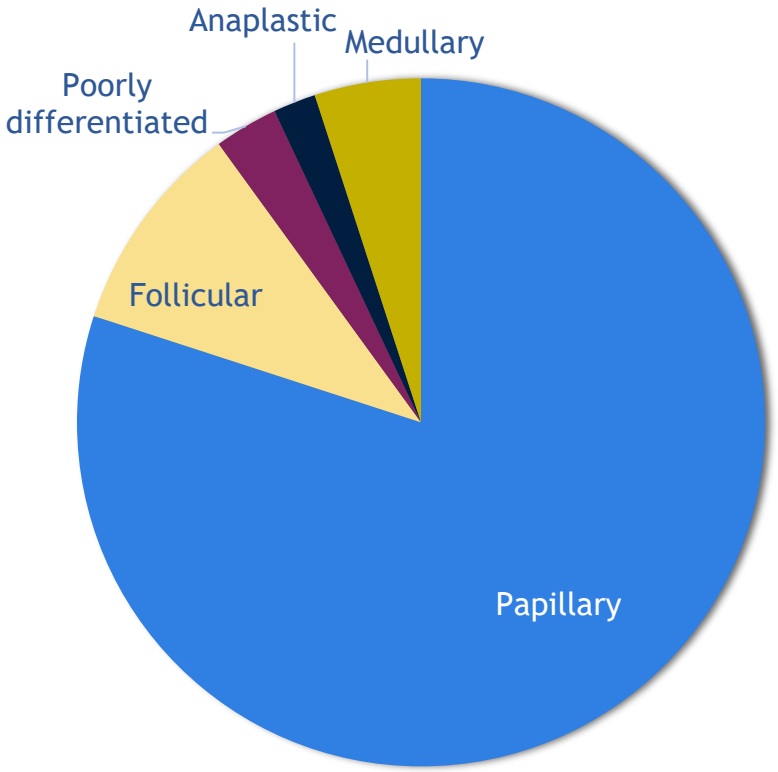


Proliferation of microfollicles, completely surrounded by a fibrous capsule



Nuclei are small, round, hyperchromatic and fairly evenly spaced

Thyroid Carcinomas



Derived from:

- Papillary
- Follicular
- Poorly differentiated
- Anaplastic

→ Follicular cells

- Medullary

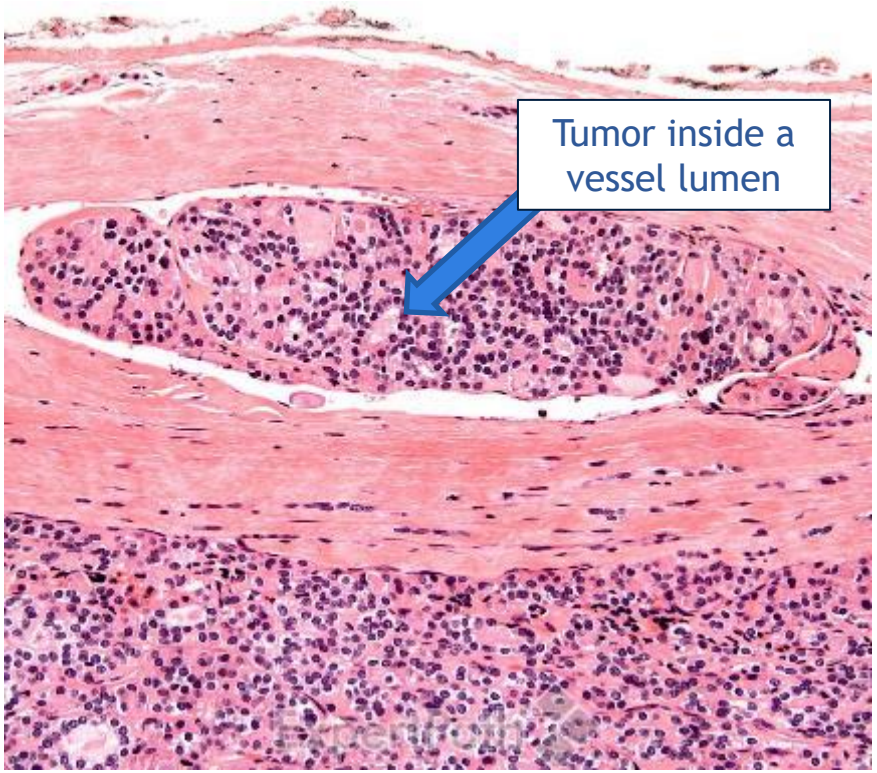
→ Parafollicular (C) cells

Follicular Thyroid Carcinoma

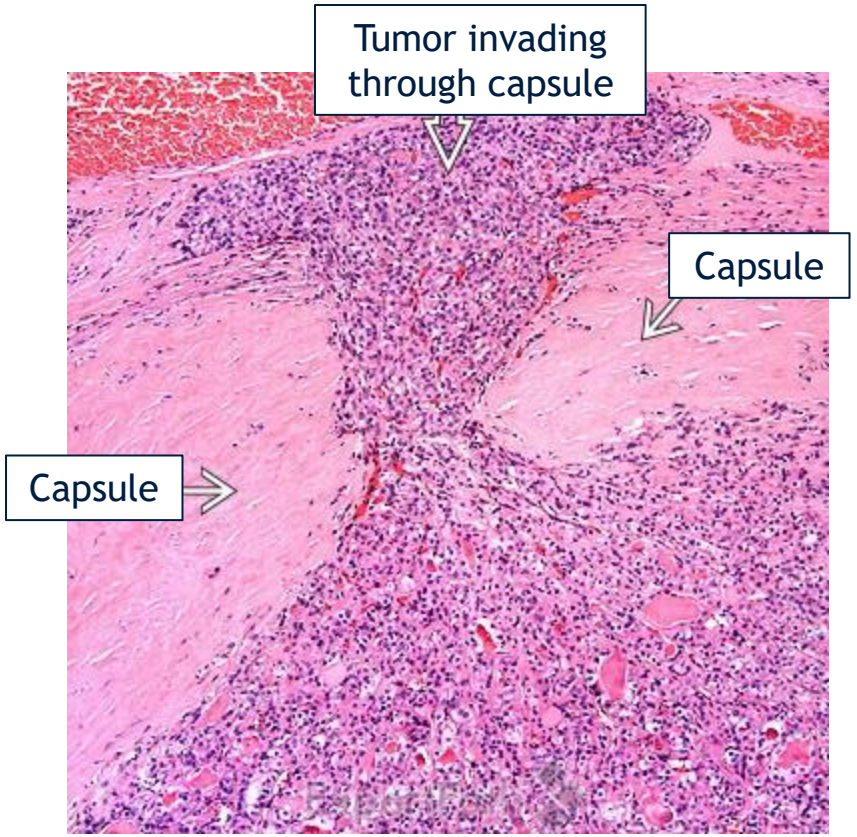
- Malignant follicular cell-derived neoplasm that has invasive growth and lacks nuclear features of papillary thyroid carcinoma
- Molecular alterations
  - *RAS* point mutations
  - *PAX8/PPARλ* fusion

Follicular Thyroid Carcinoma: Pathology

- Similar morphology (architectural & nuclear features) to follicular adenoma but has **INVASION** (vascular and/or capsular)



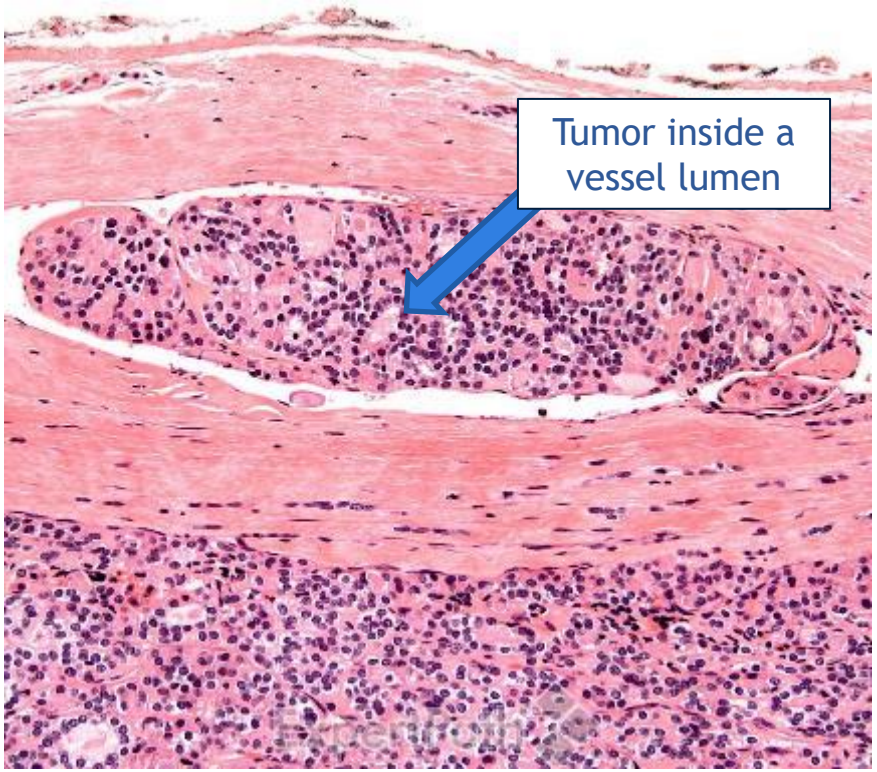
Vascular invasion



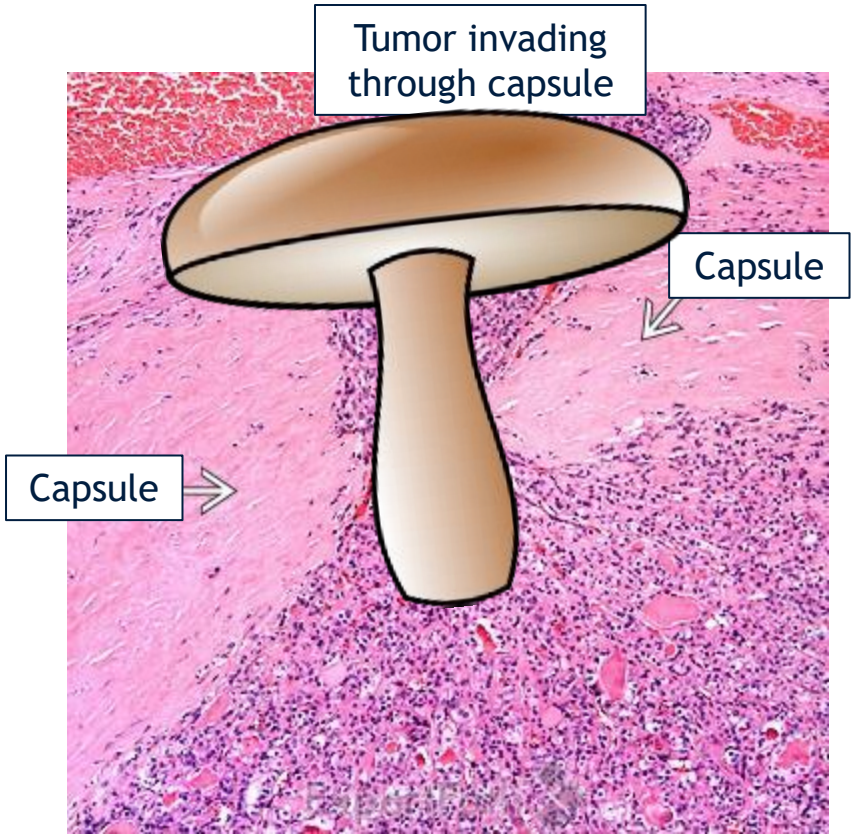
Capsular invasion ("mushroom")

Follicular Thyroid Carcinoma: Pathology

- Similar morphology (architectural & nuclear features) to follicular adenoma but has **INVASION** (vascular and/or capsular)



Vascular invasion

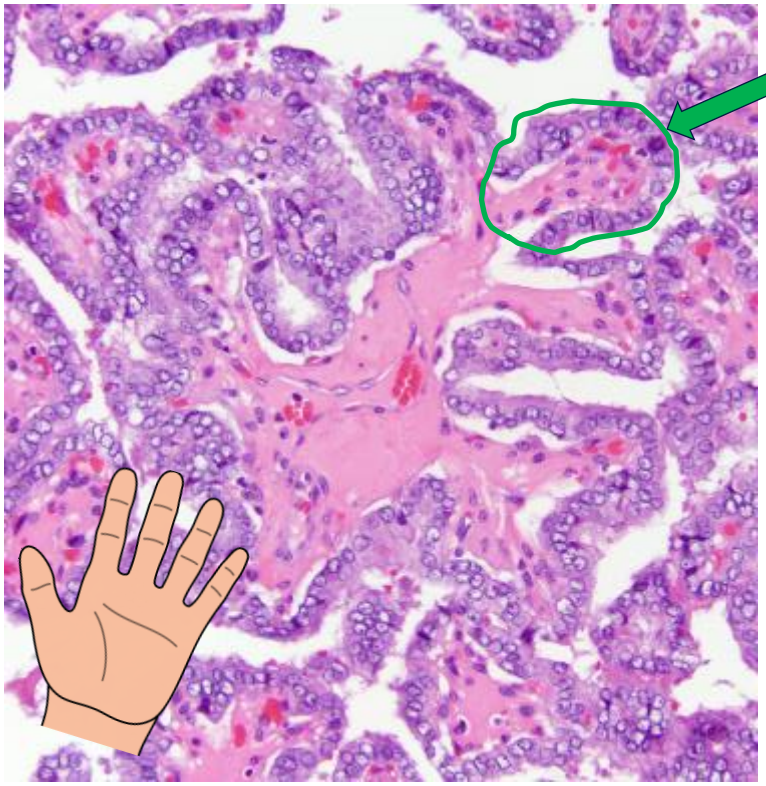


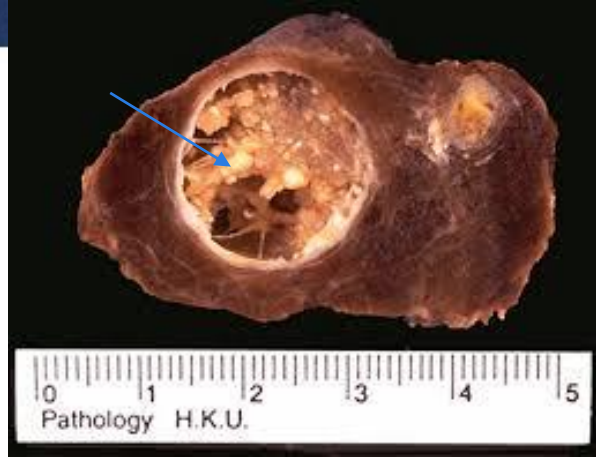
Capsular invasion ("mushroom")

Papillary Thyroid Carcinoma (PTC)

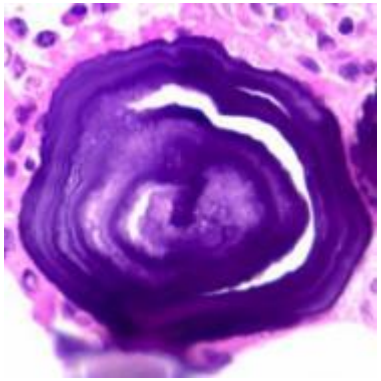
- #1 type of thyroid cancer
- Occurs at any age; female predominance
- Metastasizes via lymphatics to cervical lymph nodes
- Excellent prognosis overall (>90% 20-year survival)
  - Some subtypes have more aggressive behavior
- Most common molecular alterations
  - BRAF V600E mutation
  - RET/PTC fusion

Classic PTC: Papillary Architecture





Cystic mass with projections into the lumen



Psammoma body
= concentric lamellated calcified structure
= "ghost" of necrotic papilla

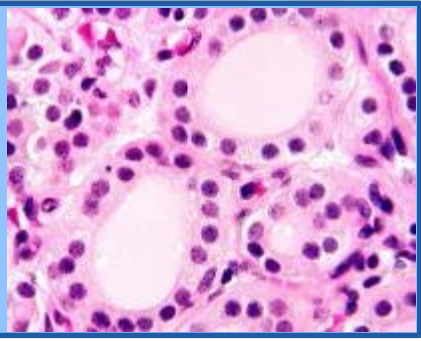
Papilla = *finger-like projection with fibrovascular core lined by tumor cells*

(plural = papillae)

fibrous tissue + *capillaries*

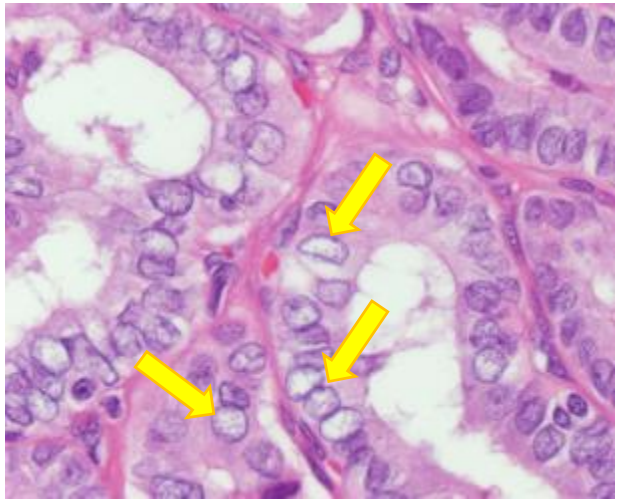
PTC: Nuclear Features

Compare to nuclear morphology of follicular adenoma/carcinoma *(small, round, hyperchromatic, evenly spaced)*:



1. Changes in size & shape

- Enlargement, overlapping, elongation



2. Chromatin changes

- Pallor
- "Clearing" with margination

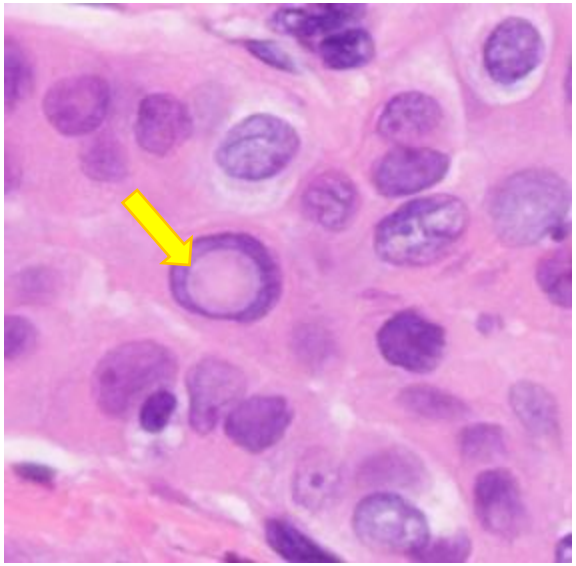


"Orphan Annie eye" nuclei

3. Nuclear membrane irregularities



Longitudinal grooves ("Coffee bean nuclei")

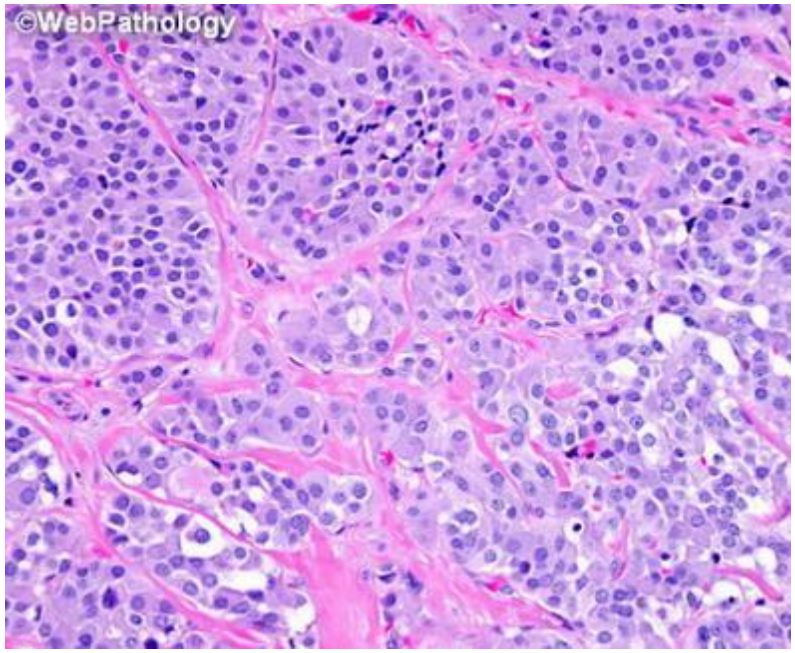


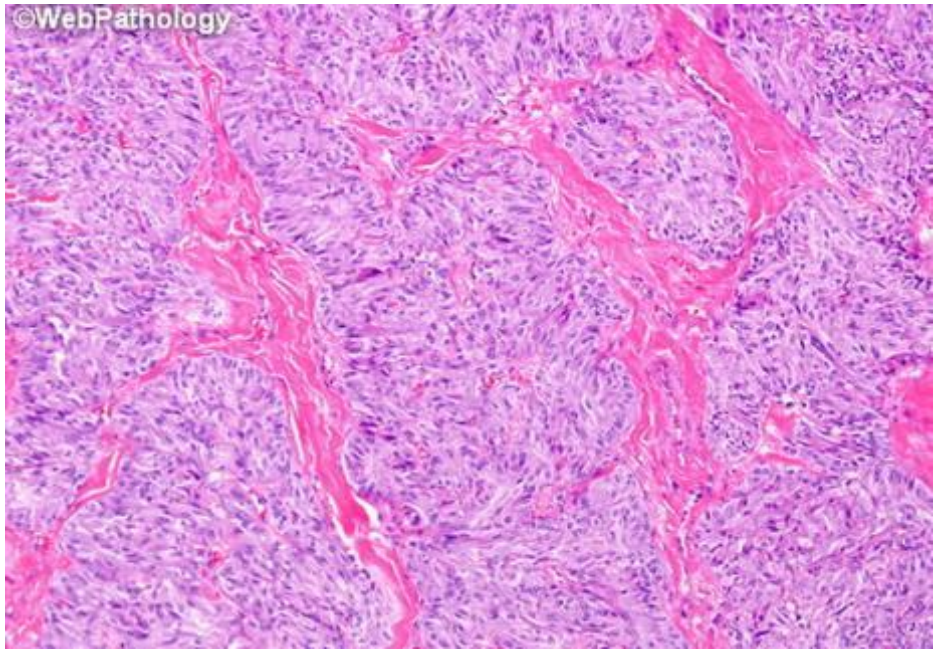
Intranuclear pseudoinclusion (invagination of cytoplasm)

Medullary Thyroid Carcinoma (MTC)

- Malignant C cell-derived tumor
- ~70% are sporadic
  - Some have somatic mutations in *RET*
- ~30% are hereditary
  - Germline mutations in *RET*
  - Multiple endocrine neoplasia type 2 (MEN2) syndromes
  - Bilateral/multifocal tumors
  - *Prophylactic total thyroidectomy indicated*

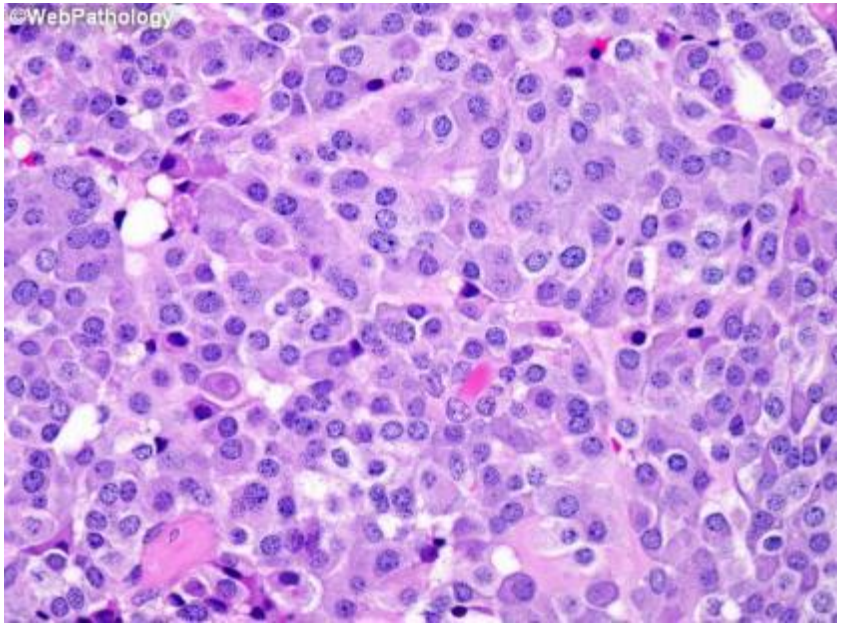
MTC: Solid, Nested or Trabecular Growth Pattern





Note: No follicles, no papillae

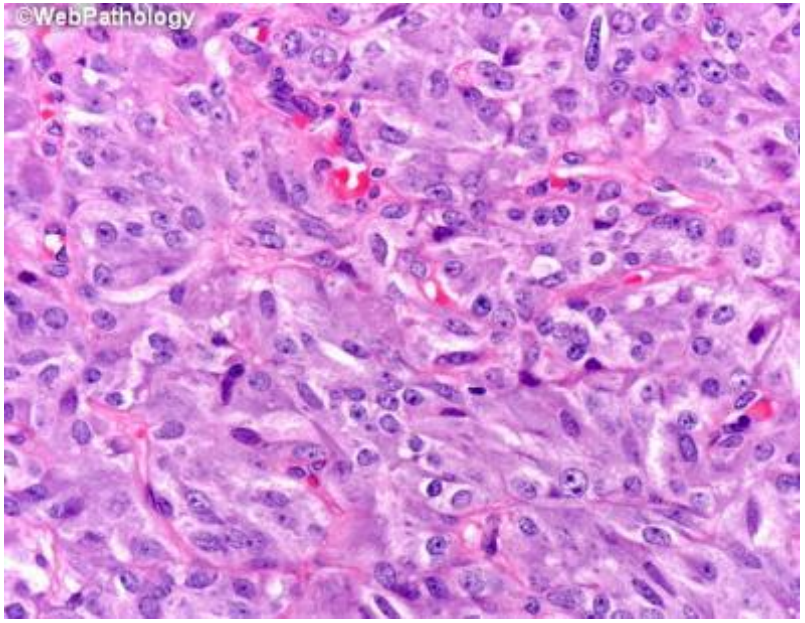
MTC: Cellular Features



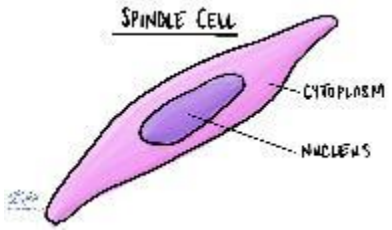
Plasmacytoid tumor cells



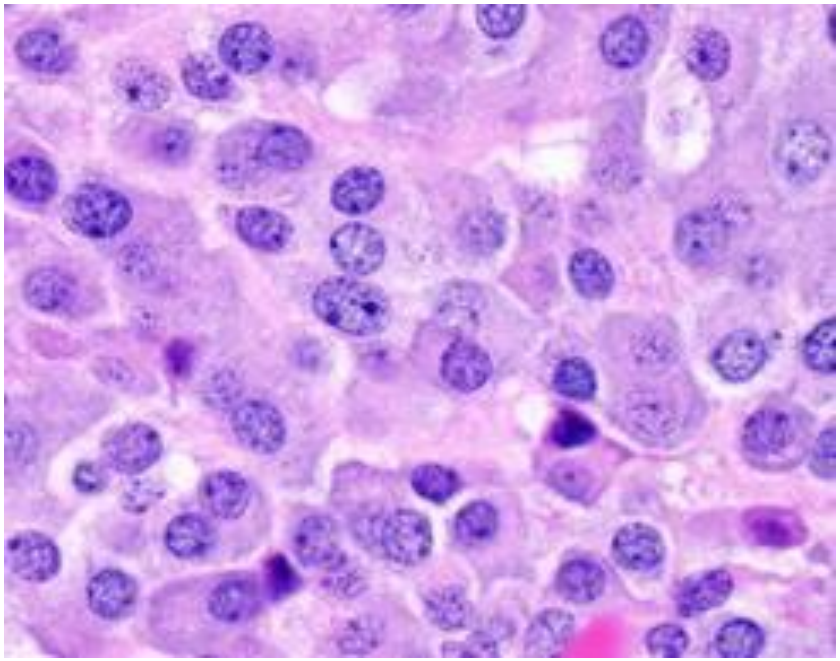
(Plasma cell: eccentric nucleus)



Spindle-shaped tumor cells



MTC: Nuclear Features



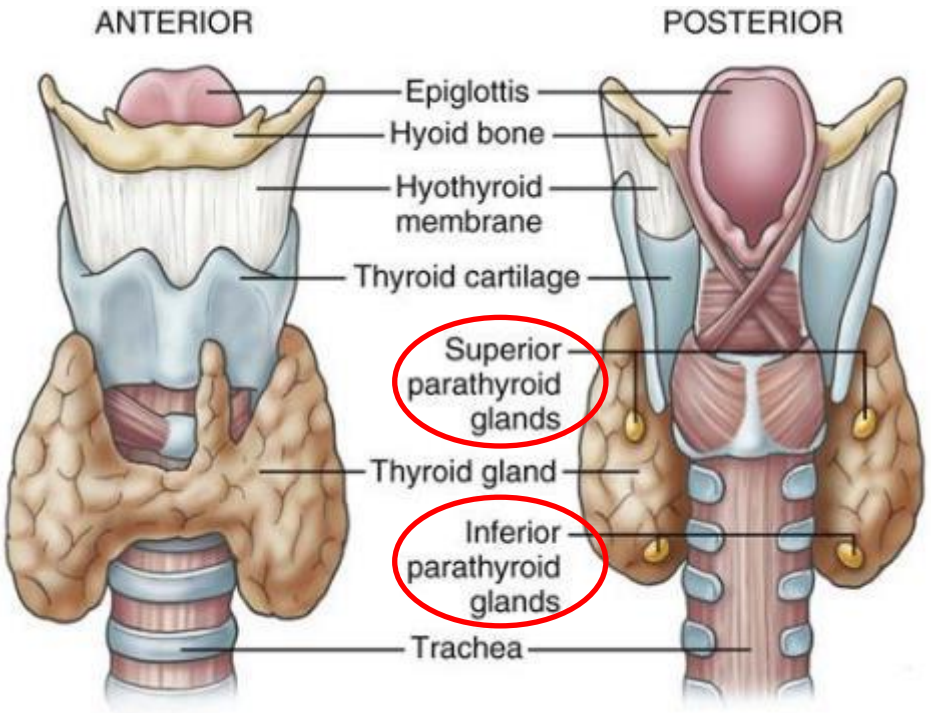
"Salt & pepper" granular chromatin



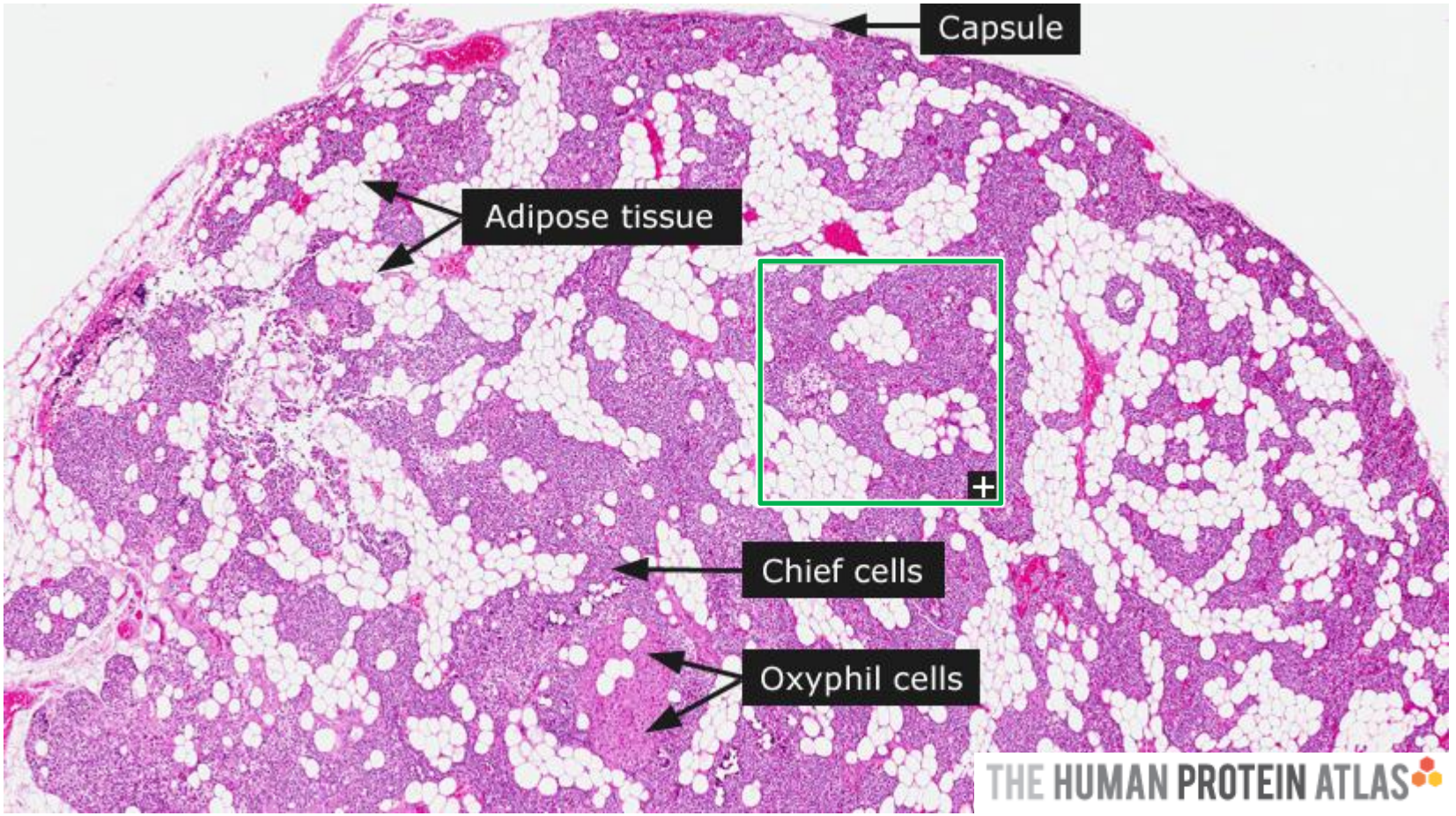
Normal Parathyroid Anatomy and Function

Gross Anatomy

- Most people (90-95%) have 4 parathyroid glands
  - Paired superior & inferior glands on posterior aspect of thyroid
- Ectopic locations
  - Intrathyroidal
  - Intrathymic
- Average normal weight = 30-40 mg

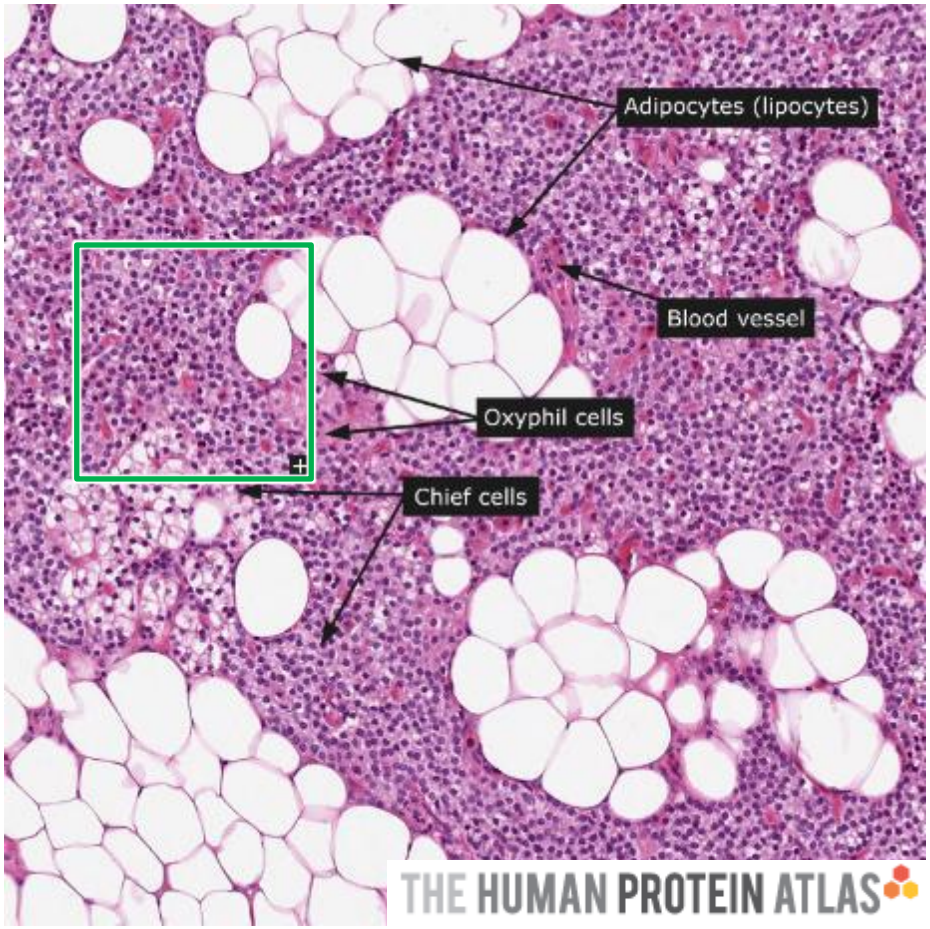


Histology (H&E)



Low power

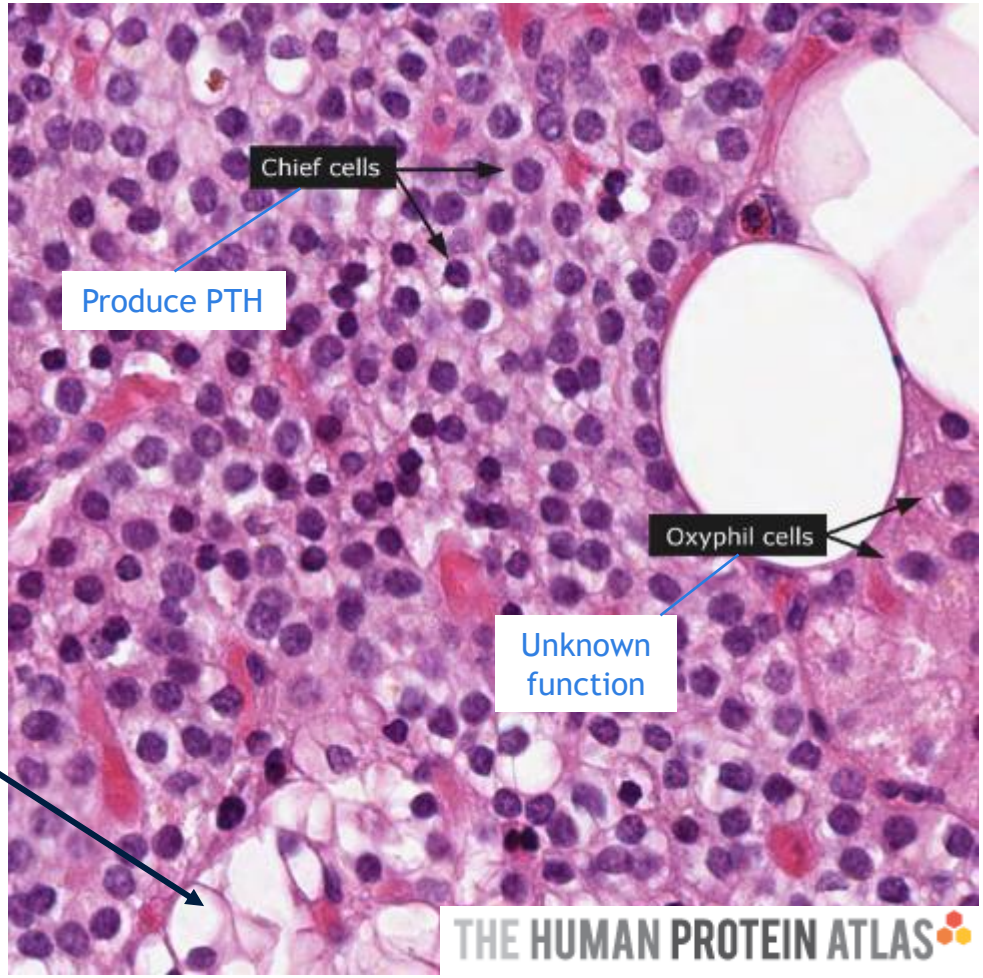
Histology (H&E)



Intermediate power

Note cellularity is approximately 50% (normal range 30-70%)

Histology (H&E)



Some chief cells have abundant clear cytoplasm (due to ↑glycogen)

Oxyphil cells are ONCOCYTIC

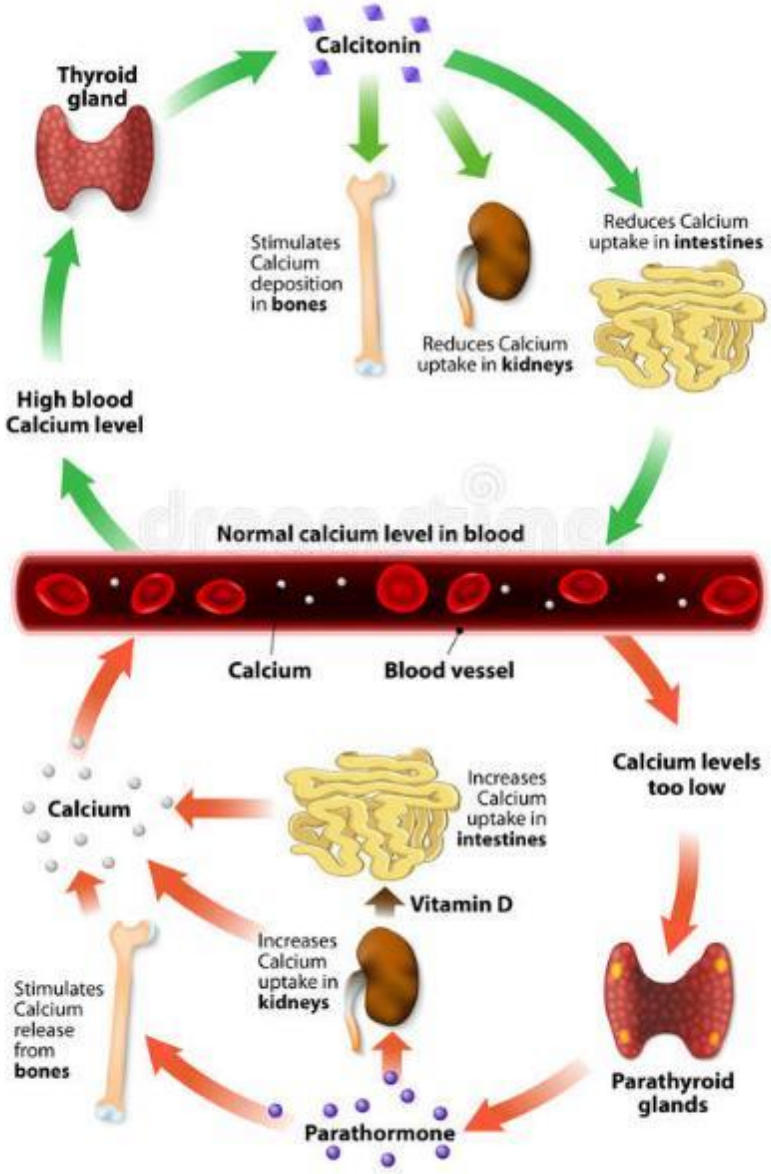
- Abundant granular, pink cytoplasm (due to ↑mitochondria)

Analogous cell type in thyroid?

- Oncocytic cells in Hashimoto thyroiditis

High power

Function



Calcium homeostasis

Hyperparathyroidism

Hyperparathyroidism: Etiology

Primary

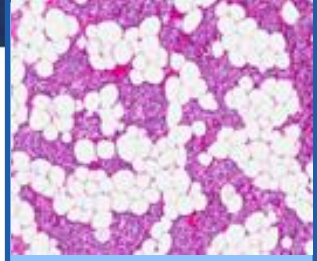
- Parathyroid adenoma (85-95%)
- Parathyroid hyperplasia (5-10%)
- Parathyroid carcinoma (1%)

#1 cause of *hypercalcemia* (→"painful bones, renal stones, abdominal groans, psychic moans")

Secondary

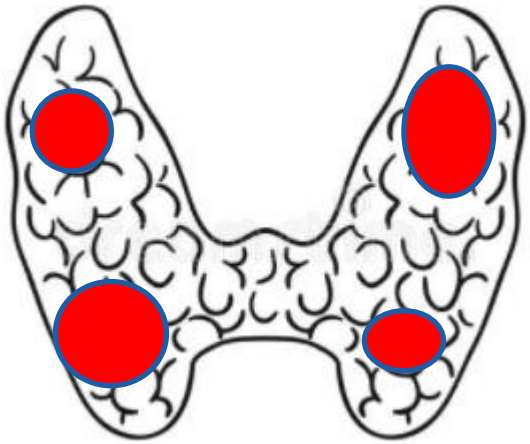
- Chronic renal failure (#1 cause)
- Vitamin D deficiency
- Malabsorption

Hypocalcemia → compensatory *hyperplasia* of parathyroid glands

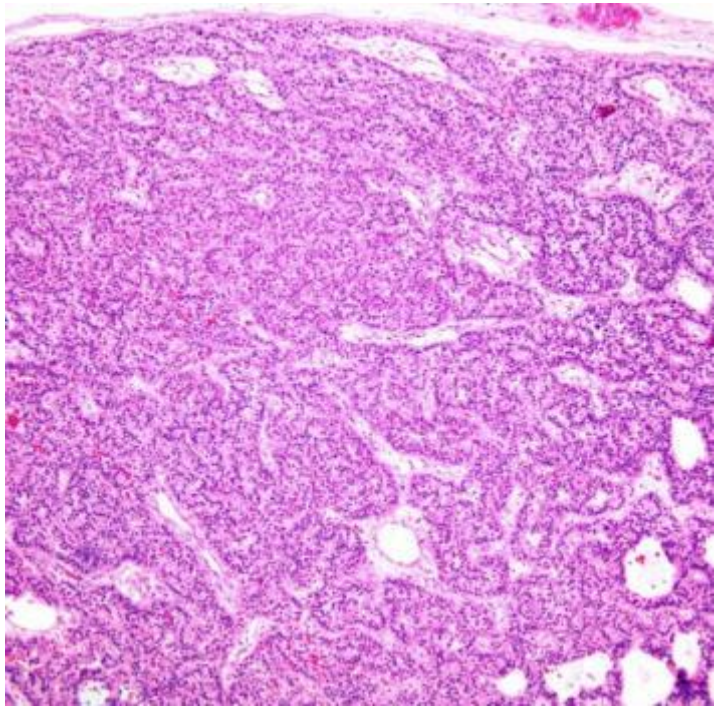

Normal

Parathyroid Hyperplasia

- Enlarged gland with ↑ cellularity (chief cells +/- oxyphil cells), ↓ fat
- Typically involves >1 gland
- 1/3 cases a/w MEN1 or MEN2A



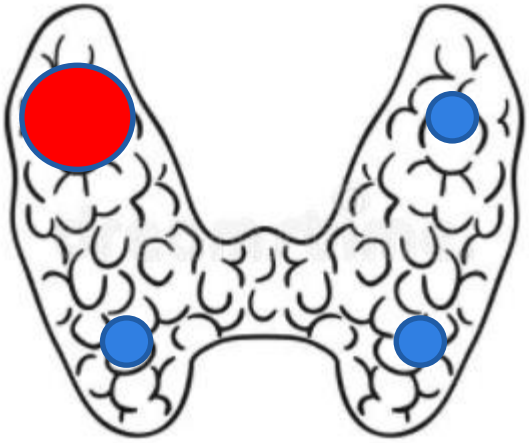
All 4 glands involved by hyperplasia



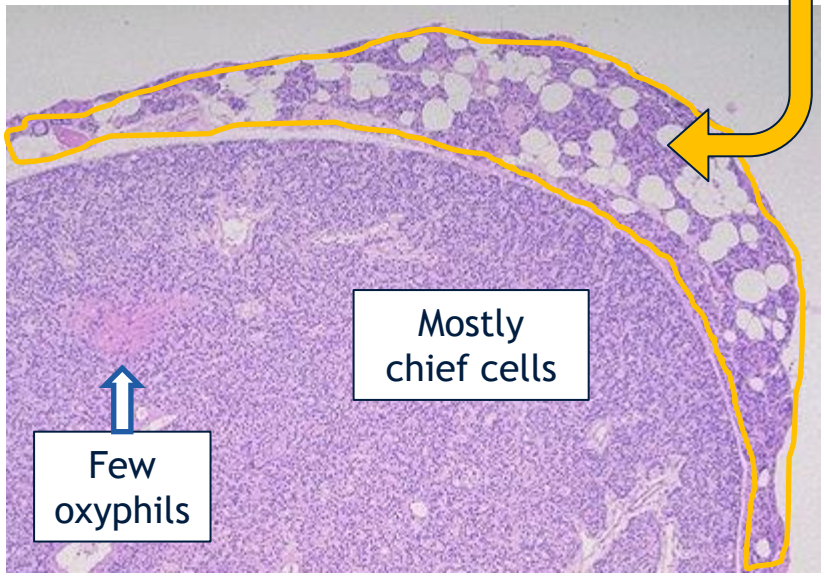
Diffuse cellular proliferation composed of chief cells without fat (100% cellular)

Parathyroid Adenoma

- Benign neoplasm of chief cells, oxyphil cells or a mixture
- Typically only 1 gland involved
- Can usually identify compressed rim of normocellular parathyroid tissue
- Sporadic or a/w MEN1/2A



1 adenoma, 3 normal glands



Nodular proliferation of cells without fat (100% cellular)

Questions?

- Stacey.Gargano@jefferson.edu


Jefferson
Philadelphia University +
Thomas Jefferson University
HOME OF SIDNEY KIMMEL MEDICAL COLLEGE